

Simple Temperature Class

Introduction

In this task, you will create a basic class to handle temperature values. This exercise will help you practice creating classes, using constructors, and implementing methods.

Class Overview

In this challenge, you will build a Temperature class that allows you to store and manipulate temperature values. The class will include methods for printing temperatures, checking if they are hot or cold, and converting between Celsius, Fahrenheit, and Kelvin.

Task

Your task is to implement the Temperature class by following these steps:

1. **Create a New Project:** Start by creating a new project with a main method.
2. **Define the Temperature Class:** Create a new class called Temperature.
 - The class should be in the same folder as the Main Class (*src*).
3. **Store Temperature in Celsius:** Create a variable to store the temperature in °C.
 - Consider the appropriate data type and name for storing values like 18.24.
4. **Default Constructor:** In the default constructor of the Temperature class, assign the value 16 to the temperature variable.
5. **Create Temperature Object:** In the main class, create a Temperature object called temp1 and print the value assigned by the constructor. Test if it works.
 - Change the value from 16 to 18.32; does it still work?
6. **Constructor Overloading:** Create another Temperature object called temp2, this time assigning the value directly during creation.
 - Hint: Use constructor overloading.
 - Test your implementation.
7. **Print Method:** Implement a method called printCelsius() in the Temperature class.
 - This method should print "n° Celsius," where n is the temperature value stored in the variable of each object created in step 3.
 - Call this method on both temp1 and temp2. Possible output:
 - 18.32° Celsius.
 - 23.67° Celsius.
8. **User Input:** Prompt the user to input a new temperature. Create a new Temperature object called temp3 and assign the user input value to it.

- How did you do it? Did you call the constructor or assign it directly?
- 9. **Test Print Method:** Call the `printCelsius()` method on the `temp3` object to test your implementation.
- 10. **Hot or Cold Check:** Create a method called `printHotOrColdCelsius()`. This method should check if `n` (the temperature) is greater than 24°.
 - If greater than 24°, print: "It's hot: n° Celsius!"
 - Otherwise, print: "It's freezing: n° Celsius!"
- 11. **Test Hot or Cold Method:** Test your newly created method with the object `temp3`. Input values lower than and higher than 24°.
- 12. **Fahrenheit Conversion:** To accommodate users in America, create a method called `calculateCelsiusToFahrenheit()` that takes a Celsius temperature value as input and calculates its Fahrenheit equivalent.
 - Consider how to name this variable; you might need to change both variable names for clarity between Celsius and Fahrenheit.
- 13. **Calculate and Print Fahrenheit:** After user input for Celsius, calculate its Fahrenheit equivalent and print it. Verify your results.
- 14. **Kelvin Conversion:** Create another method called `calculateCelsiusToKelvin()`. Implement logic similar to step 12 for converting Celsius to Kelvin.
- 15. **Test Your Application:** Make sure, that everything is working as expected.